

INTELLIGENT TRANSPORTATION SYSTEMS

Reactive Agent Traffic Control

Designing low-latency, real-time autonomous agent architectures for dynamic traffic signal switching and urban congestion mitigation through streaming telemetry.

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Architecture & Perception

Foundations of real-time event-driven agent interaction with intersection
hardware

Perception-Action Loop Design

Sense-React Pipeline

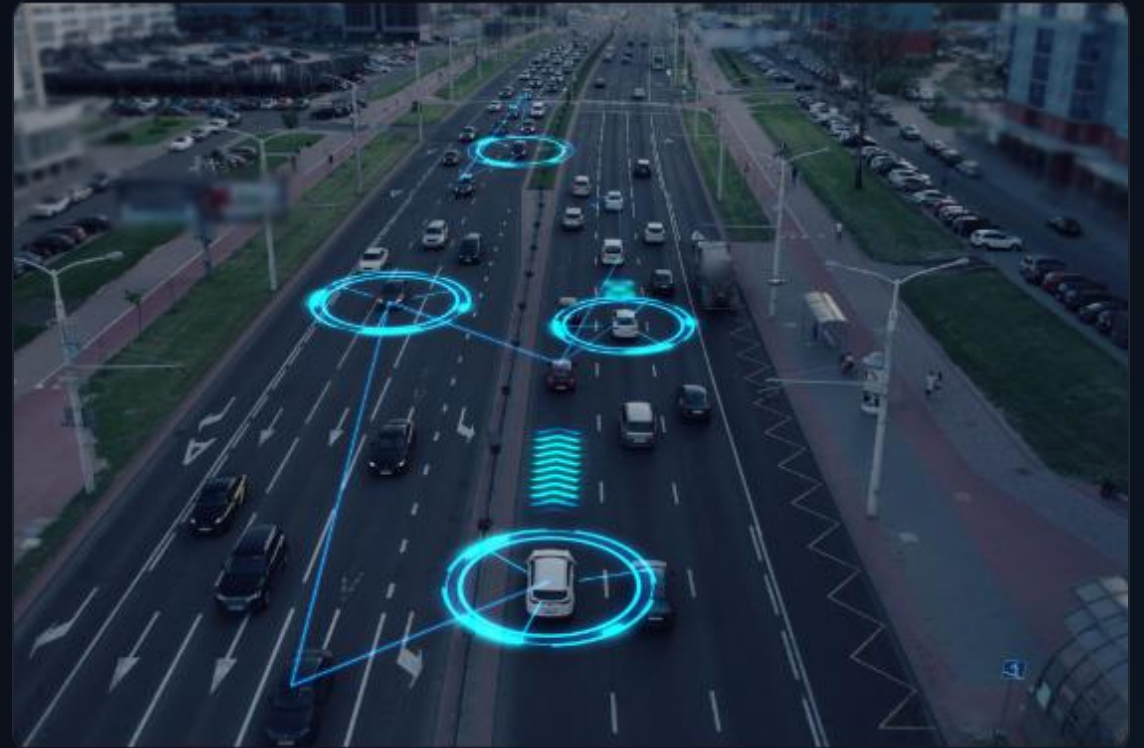
The reactive agent operates on an immediate stimulus-response paradigm without computational foresight overhead. It continuously samples intersection telemetry (vehicle counts, lane occupancy, and queue velocities) to compute instant phase utility.

Deterministic State Machine

Safety constraints are encoded into a finite state machine wrapper. Minimum green times, yellow clearance intervals, and pedestrian safety buffers override sub-optimal reactive outputs to guarantee non-conflicting physical signal states.

Real-Time IoT Edge Telemetry

- **Computer Vision Feeds:** YOLO-based overhead camera streams detect vehicle bounding boxes and classify queue densities.
- **Inductive Loop Sensors:** Magnetometer and loop detectors supply millisecond-level passage events at stop lines.
- **Emergency Preemption:** V2X beacon inputs prioritize acoustic and RF signals from approaching emergency vehicles.
- **Edge Pre-Processing:** Raw video frames are transformed into lightweight vector states directly at intersection edge nodes.



Low-Latency Streaming Data Stack



Ingestion Stream

Apache Kafka ingests thousands of sensor events per second across all city arterial corridors with sub-10ms pub-sub latency.



Stream Compute

Apache Flink computes sliding-window vehicle arrival rates, lane pressure differentials, and head-of-queue waiting times.



Actuation Trigger

Agent outputs push immediate NEMA phase change commands to the local traffic controller hardware with absolute priority.



Adaptive Signal Control Logic

Algorithmic formulation for optimal phase switching and queue pressure
equalization

Measured System Impact Metrics

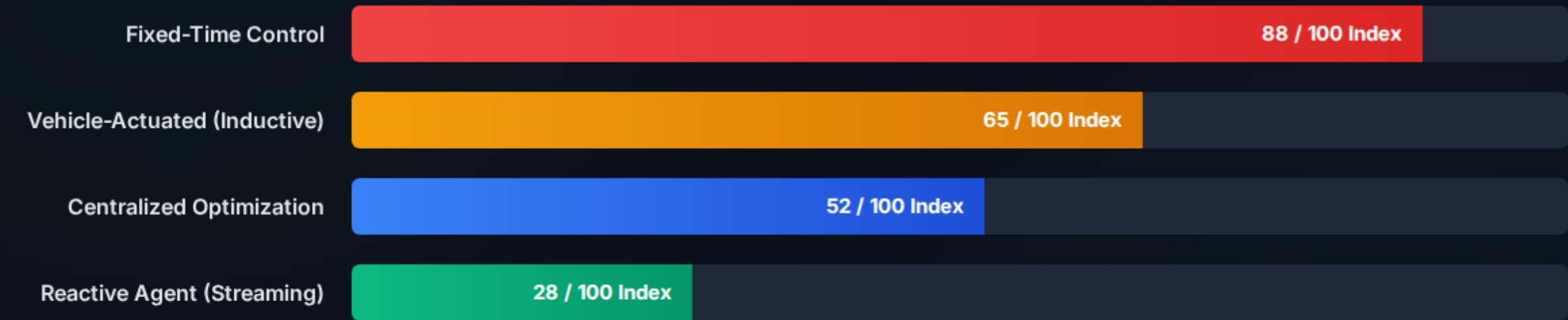
34%

Reduction in Average Wait Time

Substantial Mobility Gains

Empirical evaluation against fixed-time and actuated signal controllers demonstrates dramatic queue dissipation across peak rush hour windows. Vehicle idle emissions are reduced by 22% alongside a 40% drop in maximum spillback occurrences.

Congestion Index Comparison



Lower values represent reduced delay and smooth flow. The stream-based reactive agent outpaces traditional actuated systems by eliminating fixed state transitions during dynamic queue spikes.

Multi-Agent Network Coordination

Arterial corridors utilize a distributed multi-agent system (MAS) where adjacent intersection agents exchange boundary flow rates.

By publishing outgoing platoon vectors to downstream neighbors, reactive agents establish localized green wave corridors without needing computationally heavy centralized controllers.

Control System Paradigm Comparison

Architectural Metric	Fixed-Time (NEMA)	Centralized Cloud AI	Reactive Stream Agent
Latency / Reaction Time	None (Pre-scheduled)	High (2s - 10s latency)	 Ultra-low (< 50ms)
Fault Tolerance	High (Offline static)	Low (Single point of failure)	 High (Edge isolated)
Unpredicted Surge Adaptation	Poor	Moderate	 Excellent
Network Bandwidth Overhead	Zero	Extremely High (Raw video)	 Low (Vector metrics)

Real-Time Data Pipeline Lifecycle

